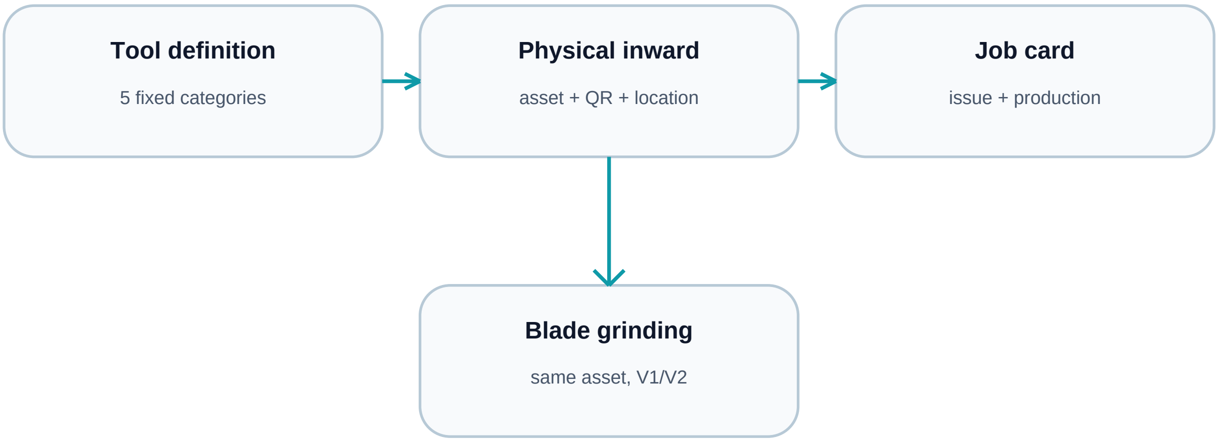
Tooling Lifecycle

Client operating report and user guide

From approved tooling master to physical QR asset, production issue, blade grinding, and trace reporting.



Release scope

This report explains the tooling and specification-sheet flows now available in the ERP. It is written for the client, store, production, quality, and supervisory teams. Technical implementation details are intentionally kept out of the operating instructions.

At a glance

Control	What the user gets
Five fixed categories	Notch, Blade, Holder, V + Flat, Punch
Physical identity	One asset number and QR value for every inwarded unit
Trace	Issue, return, maintenance, grinding, job-card output
Spec sheet	Approved values only, with mandrel-linked tube selection
Suggestions	Removed from the page and calculation flow

1. Tooling master

Open Masters > Tools. The five categories are fixed. The plant team can create multiple tool definitions under every category, but cannot create a sixth category.

Category	Fields maintained
Notch	Type, thickness, design, degree
Blade	Type, thickness, height, length
Holder	Thickness, height, length
V + Flat	Length, thickness
Punch	Punch option

Adding a definition

• Select Add Tool.

• Choose one of the five fixed categories.

• Enter the category points and status.

• Save. The definition becomes available to the specification sheet while active.

Editing approved dropdown values

Use the Editable dropdown registry to maintain the values used by the master and the notch process. Direction, distance, and depth are process option values, not extra tool categories. Only active values appear for selection.

Important

Master definition and physical asset are separate records. Updating a definition does not erase the saved snapshot or history of a tool that was already inwarded.

2. Physical inward and QR identity

Use Inward beside the tool definition. Enter receipt date, quantity, and a Location Master position. The system expands the quantity into individual physical assets.

Generated information	Example
Asset number	TA-260714-001
QR value	hariom://tool/TA-260714-001
Starting status	Available
Grinding version	V0
Location	Tool Rack A

Example: two plain blades received on 14-Jul-2026 create two separate assets. They share the same definition but each has its own location, status, issue history, usage count, and production output.

3. Physical lifecycle

The ledger shows only the actions valid for the current physical status.

Status	Use
Available	Ready to issue, maintain, grind (blade), or scrap
Issued	Assigned to a job card and stage; return after use
Maintenance	Temporarily with maintenance; complete when usable
Grinding out	Blade sent for sharpening; grinding return completes the cycle
Scrap	Permanently removed from use; history retained

Issue to production

• Search or scan the QR/asset number.

• Select Issue.

• Enter the job card ID and production stage.

• Save. The asset becomes Issued and the job card stores the exact asset ID.

Blade grinding

Return the blade if it is issued, select Grinding out, send it for sharpening, then select Grinding return. The same asset number remains in use and the version increments from V0 to V1, then V2, and so on.

A non-blade tool cannot use the blade grinding action.

4. Production output and trace

At job-card completion, actual completed output is recorded against the physical asset IDs assigned to the stage. A retry cannot double-count the same completion.

Trace point	Example
Job card	JC-2026-014
Stage	Notching
Physical asset	TA-260714-001
Good output	1,200 tubes
Scrap output	15 tubes
Next lifecycle	Returned, then grinding V1

This answers how much each blade produced, how much scrap was associated with it, which job cards used it, and how many grinding cycles it completed.

5. Specification sheet

Open Specifications > New Specification. The sheet uses active tool definitions and active process option values. Discontinued records are not shown.

The notch section contains exactly eight process fields

- Notch type
- Notching blade
- Notching holder
- V + Flat
- Punch
- Notch direction
- Notch distance
- Notch depth

The first five are tool selections. Direction, distance, and depth are editable process options. Tool points are displayed from the master and are not retyped in the specification sheet.

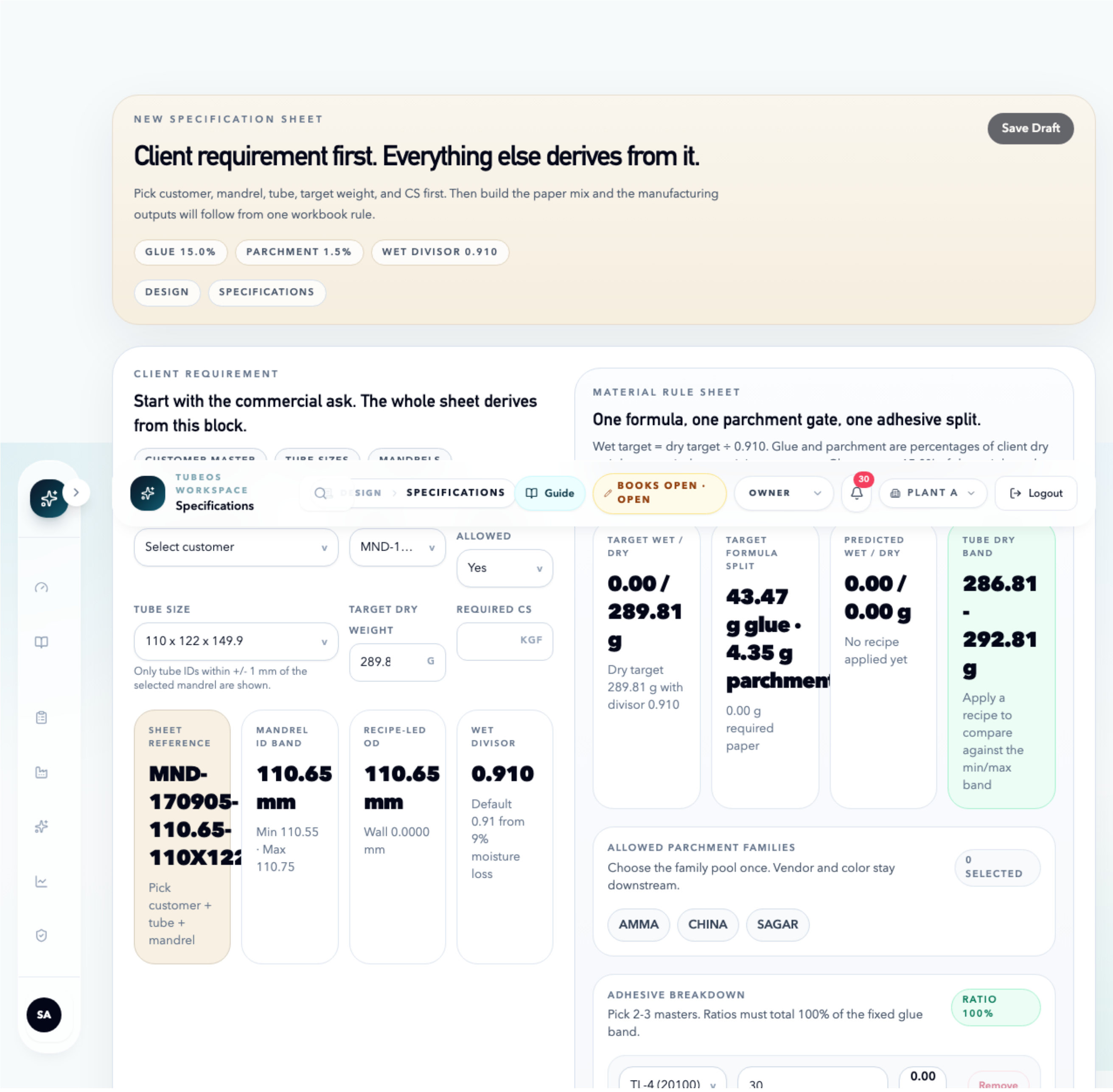
Mandrel and tube rule

Select a mandrel first. The tube-size picker then becomes active and shows only tube IDs within plus or minus 1 mm of the selected mandrel. This keeps the sheet aligned with the physical mandrel choice.

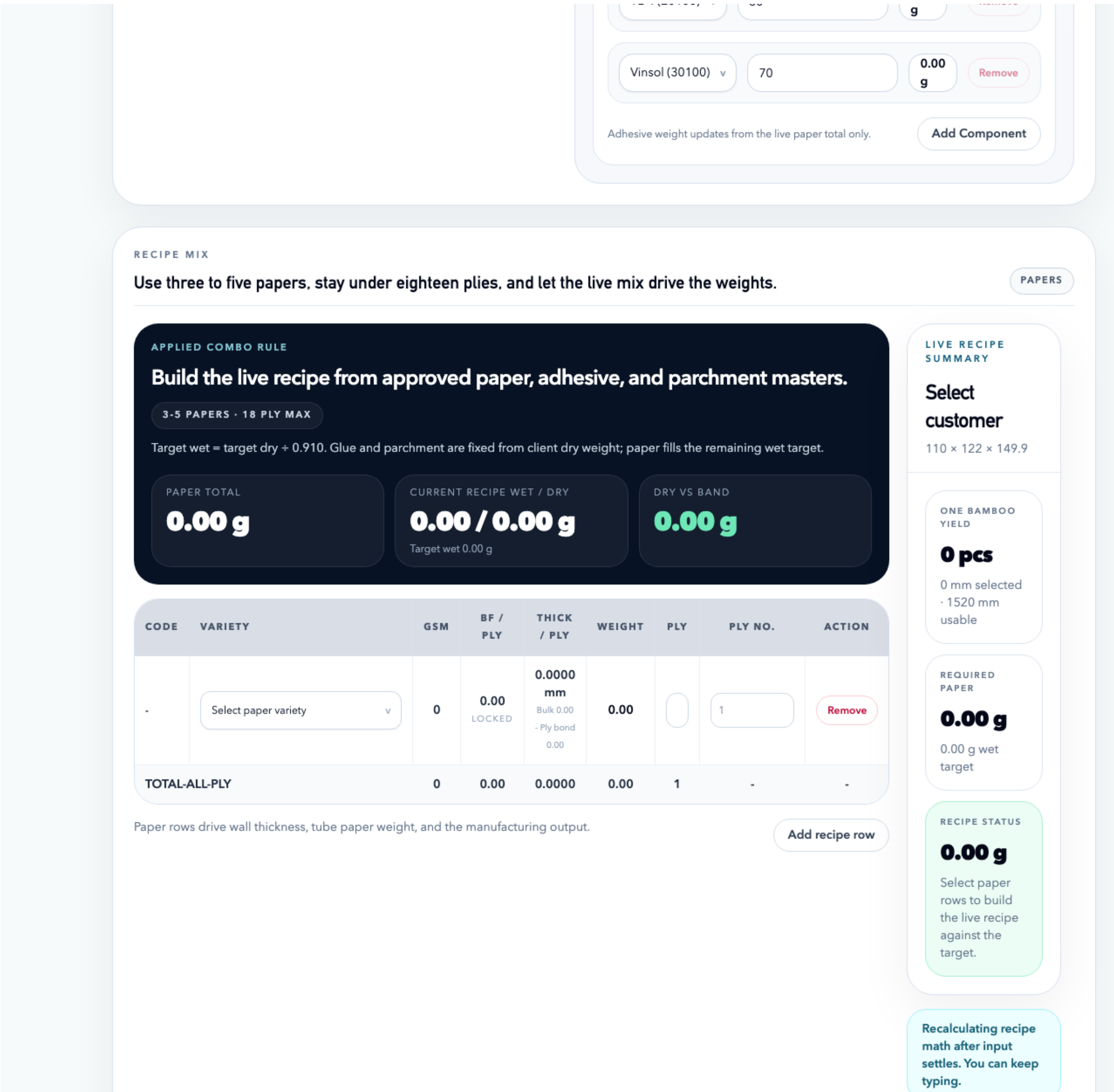
Suggestions are removed

The specification sheet no longer shows suggestion cards or runs the removed suggestion calculation route. The user builds and saves the approved recipe directly from the masters and entered production values.

Browser evidence - searchable mandrel, filtered tube control, and no suggestion card (1 of 3)


NEW SPECIFICATION SHEET
Save Draft
Client requirement first. Everything else derives from it.
Pick customer, mandrel, tube, target weight, and CS first. Then build the paper mix and the manufacturing outputs will follow from one workbook rule.
GLUE 15.0%
PARCHMENT 1.5%
WET DIVISOR 0.910
DESIGN
SPECIFICATIONS
CLIENT REQUIREMENT
Start with the commercial ask. The whole sheet derives from this block.
MATERIAL RULE SHEET
One formula, one parchment gate, one adhesive split.
Wet target = dry target ÷ 0.910. Glue and parchment are percentages of client dry
TUBEOS WORKSPACE
Specifications
SPECIFICATIONS
Guide
BOOKS OPEN · OPEN
OWNER
30
PLANT A
Logout
Select customer
MND-1...
ALLOWED
Yes
TUBE SIZE
110 x 122 x 149.9
Only tube IDs within +/- 1 mm of the selected mandrel are shown.
TARGET DRY WEIGHT
289.8
G
REQUIRED CS
KGF
SHEET REFERENCE
MND-170905-110.65-110X122
Pick customer + tube + mandrel
MANDREL ID BAND
110.65 mm
Min 110.55 · Max 110.75
RECIPE-LED OD
110.65 mm
Wall 0.0000 mm
WET DIVISOR
0.910
Default 0.91 from 9% moisture loss
TARGET WET / DRY
0.00 / 289.81 g
Dry target 289.81 g with divisor 0.910
TARGET FORMULA SPLIT
43.47 g glue · 4.35 g parchment
0.00 g required paper
PREDICTED WET / DRY
0.00 / 0.00 g
No recipe applied yet
TUBE DRY BAND
286.81 - 292.81 g
Apply a recipe to compare against the min/max band
ALLOWED PARCHMENT FAMILIES
Choose the family pool once. Vendor and color stay downstream.
0 SELECTED
AMMA
CHINA
SAGAR
ADHESIVE BREAKDOWN
Pick 2-3 masters. Ratios must total 100% of the fixed glue band.
RATIO 100%
30
0.00
Remove
SA

Browser evidence - searchable mandrel, filtered tube control, and no suggestion card (2 of 3)



Browser evidence - searchable mandrel, filtered tube control, and no suggestion card (3 of 3)

MANUFACTURING MATRIX

ID comes from mandrel, OD comes from wall, and bamboo output follows the live recipe.

Bamboo weight is now the predicted recipe output for the usable cut length after trim loss.

MANDREL ID BAND

110.65 mm

Min 110.55 · Max 110.75

WALL / OD

0.0000 / 110.65 mm

OD = ID + 2 × wall

CURRENT RECIPE TUBE

0.00 / 0.00 g

Target wet 0.00 g · Paper 0.00 g · Glue 0.00 g

BAMBOO WET / DRY

0.00 / 0.00 g

0 pcs from 1520 mm usable

MANUFACTURING SPECIFICATION

Average is the live working size. Min/max stay tied to mandrel tolerance and recipe wall.

PARAMETER	MIN	AVG	MAX
I.D	110.55 mm	**110.65 mm**	110.75 mm
O.D	110.55 mm	**110.65 mm**	110.75 mm
Thickness	0.00 mm	**0.00 mm**	0.00 mm
Bamboo LT	0 mm	**0 mm**	10 mm
Tube Dry	287 g	**290 g**	293 g
Tube Wet	315 g	**0 g**	322 g

BAMBOO AND OUTPUT LOGIC

This block is what moves downstream into the job card.

SELECTED BAMBOO

0 mm

1520 mm usable after trim

TUBES / BAMBOO

0 pcs

150 mm tube length

PAPER / MM

0.0000 g

0.00 g required paper

NOTCH + TOOLING

Only the downstream tooling fields that still matter to the job card and print sheet.

EXPAND WHEN NEEDED

PACKING

Compact packing handoff only: box, plastic, fadda, and per-box counts.

EXPAND WHEN NEEDED

VALIDATION

Final footer and release checks only. Keep this out of the way until the sheet is ready.

EXPAND WHEN NEEDED

6. Reports and rejection trace

Open Reports > Tooling. Use the report for current physical status and output by asset.

- Total assets and counts by status.
- Category-level physical stock.
- Searchable physical ledger with location and QR value.
- Usage count and produced quantity per asset.
- Scrap quantity, grinding version, and current job assignment.
- Lifecycle history for inward, issue, return, maintenance, grinding, and scrap.

Customer rejection investigation

- Start with the order, finished-good lot, or job card.
- Read the physical tool asset IDs on the completed job-card stage.
- Open each asset from the tooling ledger.
- Review inward, location, issue/return, maintenance, grinding versions, and production output.
- Compare the asset output with the job-card and QC records.

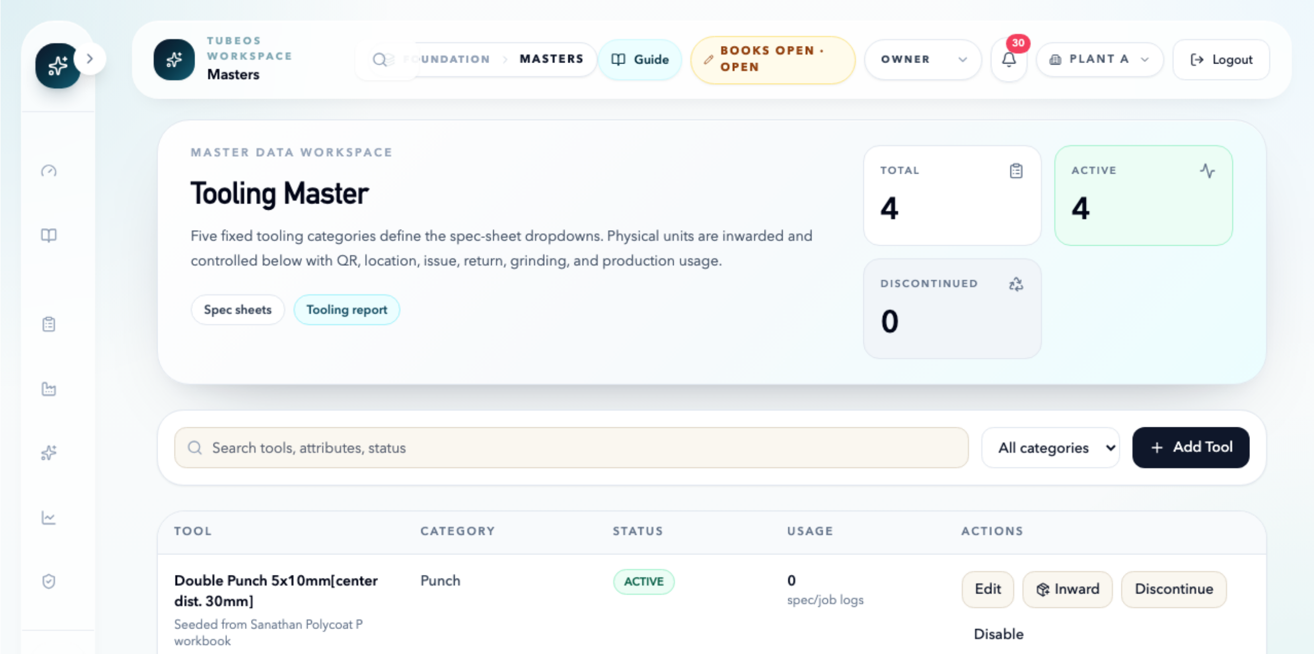
The result is a forward and backward trace from customer rejection to job card, physical tool, inward record, and maintenance/grinding history.

7. Daily checklist

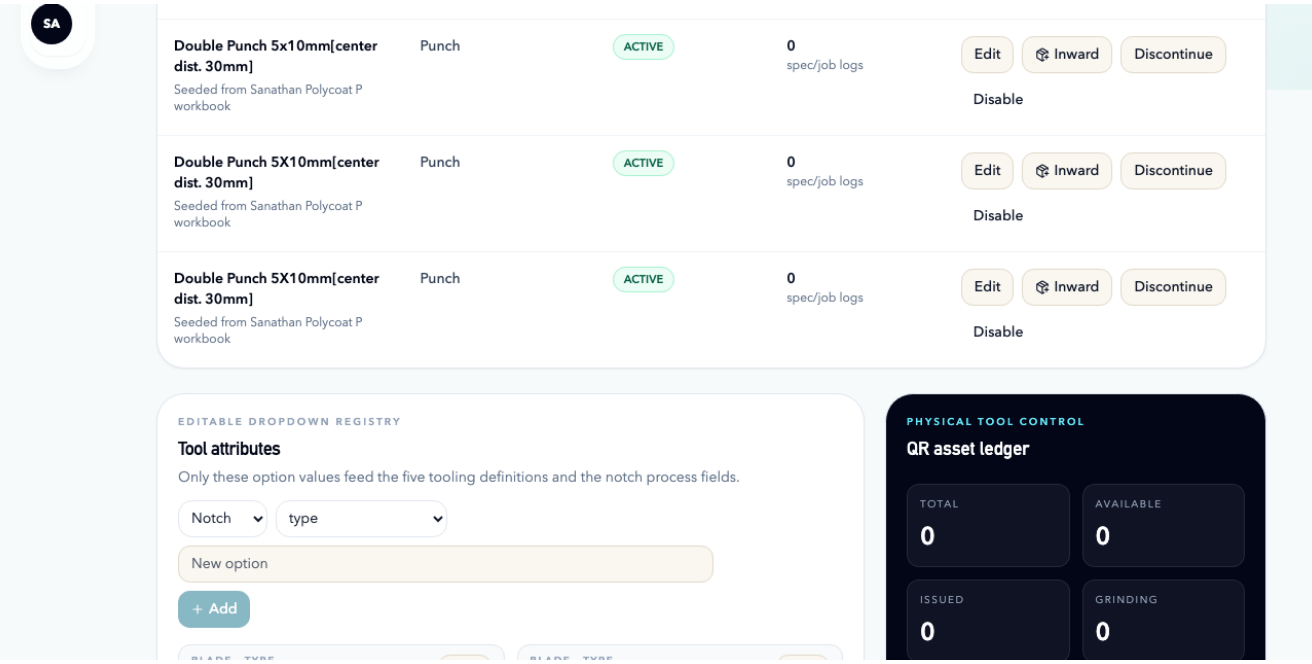
- Maintain definitions and approved dropdown values before use.
- Inward every physical unit and assign its Location Master position.
- Apply the QR label.
- Issue and return the exact asset on every job card.
- Use grinding only for blades.
- Complete the job card so output is recorded.
- Review the tooling report at shift or month end.
- Discontinue or scrap records instead of deleting history.

Browser evidence - tooling master, option registry, and QR ledger

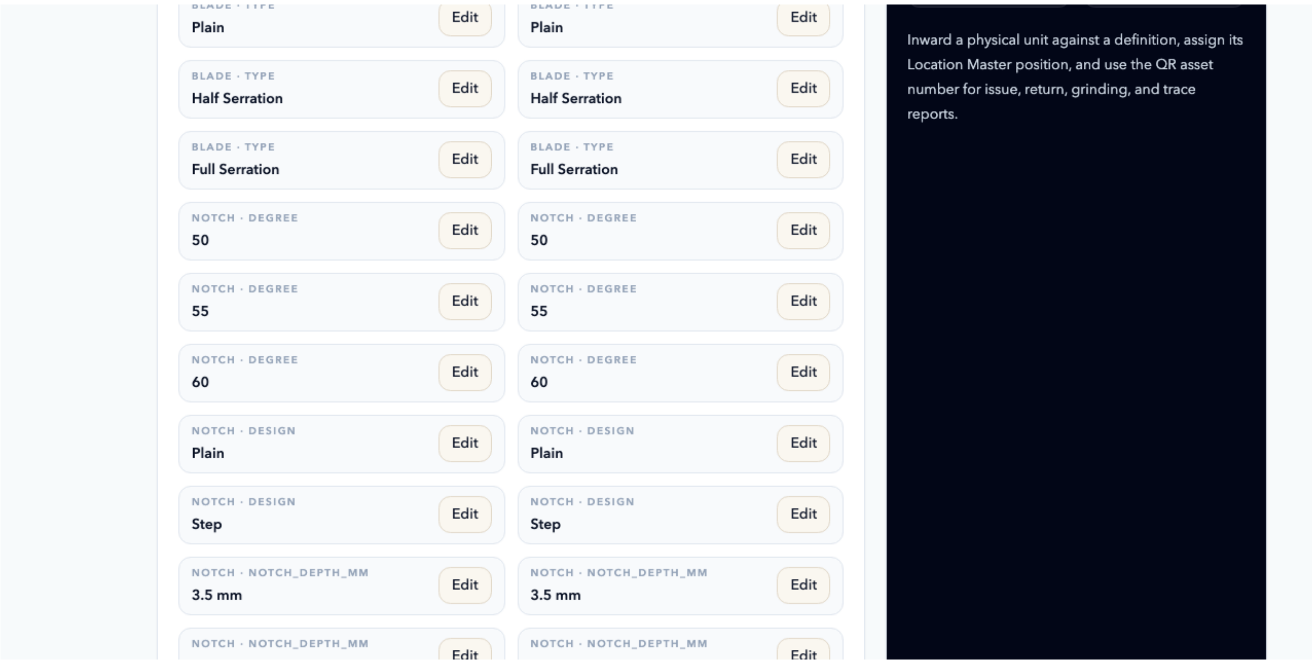
Tooling master and physical ledger (1 of 5)



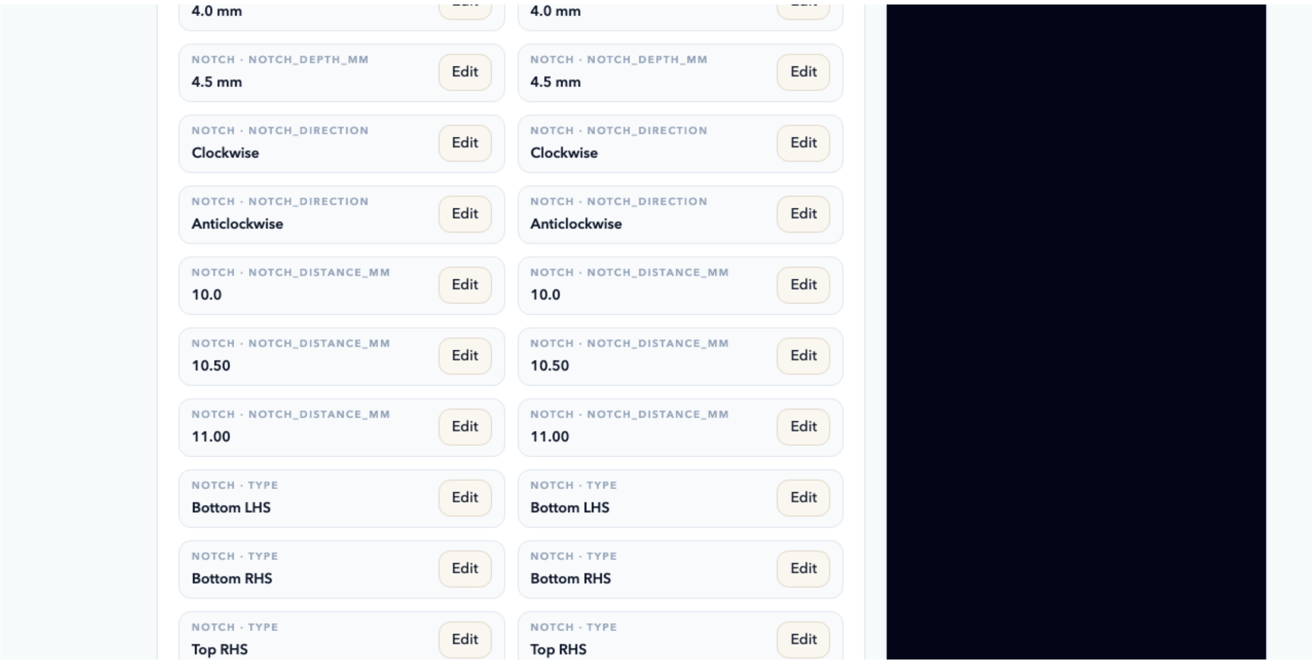
Tooling master and physical ledger (2 of 5)



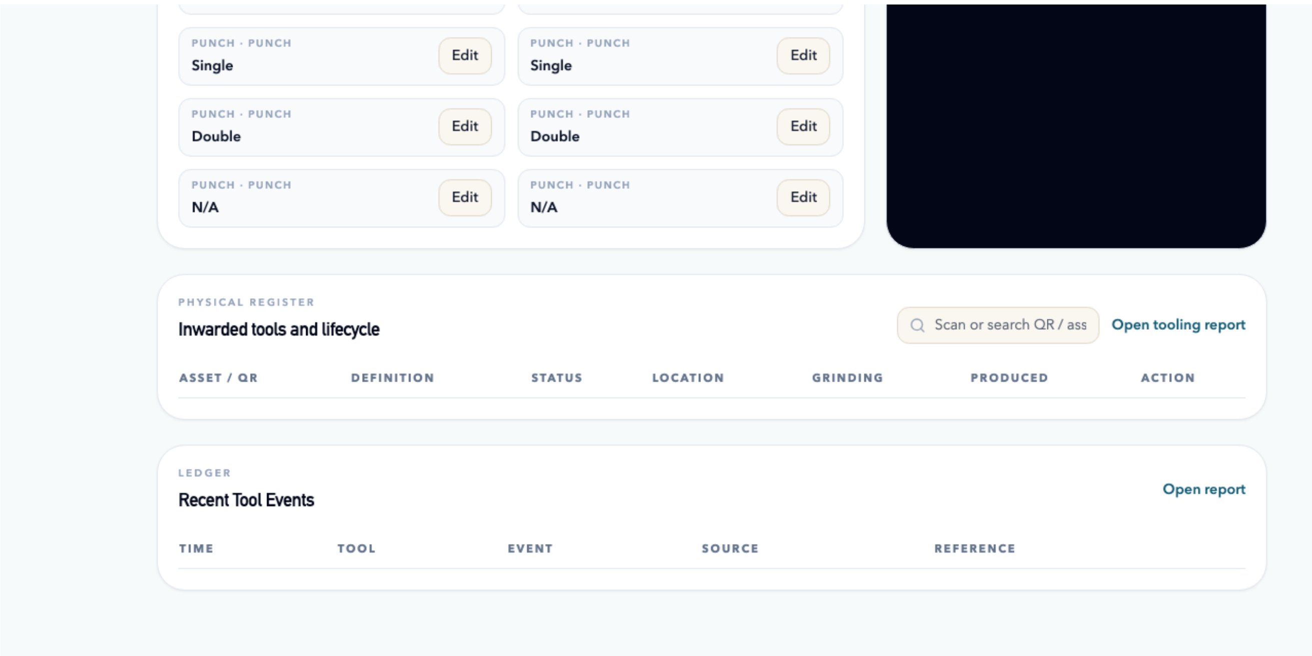
Tooling master and physical ledger (3 of 5)



Tooling master and physical ledger (4 of 5)



Tooling master and physical ledger (5 of 5)



Release verification

The implementation was checked locally with service, unit, build, static, and focused browser verification. The application is ready for client data onboarding; the final plant acceptance steps are listed below.

Deployment status

Release commit 8740c55 is committed and pushed to GitHub. The linked Railway service remains online on its previous deployment. Railway refused the new upload because the account trial has expired; after selecting a Railway plan, redeploy this commit and run the authenticated production smoke test.

Verification	Result
Service compilation	Passed
Master tooling contract	3 passed
Physical lifecycle tests	2 passed
Web unit/static checks	Passed
TypeScript and production build	Passed
Dependency audit	0 vulnerabilities
Focused browser tooling/spec suite	2 passed
BFF health and service startup	Healthy / ready

Client onboarding acceptance

- Confirm Location Master positions.
- Load approved tool definitions and option values.
- Inward opening physical tool stock.
- Print and apply QR labels.
- Run one supervised issue/return cycle.
- Run one supervised blade grinding cycle.
- Complete one supervised job card and verify the tooling report.

Prepared for client operating use